

AP CALCULUS AB – UNIT 5

Analytical Applications of Differentiation

Comprehensive Coloured Review



Name: _____ Class: _____ Date: _____

Review Scope and Learning Objectives

- Verify hypotheses and apply Rolle's Theorem and the Mean Value Theorem.
- Use first-derivative information to determine intervals of increase/decrease and relative extrema.
- Use second-derivative information to determine concavity and points of inflection.
- Connect the graphs of a function and its derivatives and justify global conclusions.
- Construct and solve optimization models with domain and endpoint analysis.

Website Lesson Map

Lesson	Website title	Review target
5.1	Mean Value Theorem	Use derivative evidence to justify analytical conclusions and optimize.
5.2-5.5 I	Derivative Graph Analysis - Part 1	Use derivative evidence to justify analytical conclusions and optimize.
5.2-5.5 II	Derivative Graph Analysis - Part 2: Challenge Questions	Use derivative evidence to justify analytical conclusions and optimize.
5.10	Optimization Problems	Use derivative evidence to justify analytical conclusions and optimize.

Key Knowledge, Formulas, and Theorems

- MVT: if f is continuous on $[a, b]$ and differentiable on (a, b) , then $f'(c) = \frac{f(b) - f(a)}{b - a}$ for some c .
- $f' > 0$ means increasing; $f' < 0$ means decreasing; sign changes classify relative extrema.
- $f'' > 0$ means concave up; $f'' < 0$ means concave down; an inflection point requires a concavity change.
- Absolute extrema on a closed interval occur among endpoints and interior critical points.
- Optimization workflow: define variables, write the objective, reduce to one variable, restrict the domain, test candidates, interpret.

Strategy and Precision Checklist

1. Identify the representation and the exact quantity requested before calculating.
2. Write the relevant definition, theorem, or derivative rule before substitution.
3. Preserve exact values in non-calculator work; use radian mode and appropriate rounding in calculator work.
4. For contextual conclusions, include the input, sign, units, and meaning.
5. Justifications must state why a theorem or procedure applies, not merely name it.

Solved Mixed Examples

Solved Example: Example 1 – 5.2-5.5 I

If $f'(x) = (x + 2)(x - 1)$, on which interval is f decreasing?

Answer and Reasoning

Result: $(-2, 1)$. A product of two linear factors with positive leading coefficient is negative between its roots.

Solved Example: Example 2 – 5.2-5.5 I

Suppose f' changes from positive to negative at $x = 3$. What occurs there?

Answer and Reasoning

Result: A relative maximum of f . The function changes from increasing to decreasing.

Solved Example: Example 3 – 5.2-5.5 II

If $f''(x) > 0$ on an interval, which statement is true?

Answer and Reasoning

Result: f is concave up and f' is increasing. A positive second derivative means tangent slopes increase, which is concave up.

Solved Example: Example 4 – 5.2-5.5 II

Find the inflection point x -coordinate of $f(x) = x^3 - 6x$.

Answer and Reasoning

Result: 0. $f''(x) = 6x$ changes sign at zero.

Leveled Mixed Practice

Foundation

Questions in this band integrate the website lessons at an increasingly demanding level.

1. Which hypotheses are required by the Mean Value Theorem on $[a, b]$?

2. For $f(x) = x^3$ on $[0, 5]$, find a value c guaranteed by the Mean Value Theorem.

3. If $f'(x) = (x + 2)(x - 1)$, on which interval is f decreasing?

4. Suppose f' changes from positive to negative at $x = 3$. What occurs there?

Core

Questions in this band integrate the website lessons at an increasingly demanding level.

5. If $f''(x) > 0$ on an interval, which statement is true?

6. Find the inflection point x -coordinate of $f(x) = x^3 - 6x$.

7. Which set must be checked to find an absolute maximum of a continuous function on a closed interval?

8. A rectangle has perimeter 58. If its width is x , which area model uses one variable?

Advanced

Questions in this band integrate the website lessons at an increasingly demanding level.

9. What dimensions maximize the area of a rectangle with perimeter 32?

10. At a point where $f'(c) = 0$ and $f''(c) < 0$, the second derivative test concludes

11. Rolle's Theorem is a special case of the Mean Value Theorem when

12. Which hypotheses are required by the Mean Value Theorem on $[a, b]$?

Difficult

Questions in this band integrate the website lessons at an increasingly demanding level.

13. For $f(x) = x^3$ on $[0, 3]$, find a value c guaranteed by the Mean Value Theorem.

14. If $f'(x) = (x + 2)(x - 4)$, on which interval is f decreasing?

15. A cylinder has radius 5.81 and height 2.5. Which value is its volume to three decimals?

16. For $f(x) = x^3 - 3x^2 + 2$, use a calculator to find $f''(1.84)$.

Challenge

Questions in this band integrate the website lessons at an increasingly demanding level.

17. A calculator optimization of $A(x) = x(11.35 - x)$ gives which maximum area?

1. At a point where $f'(c) = 0$ and $f''(c) < 0$, the second derivative test concludes

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5. If $f'(x) = (x + 2)(x - 2)$, on which interval is f decreasing?

- (A) $(-\infty, -2) \cup (2, \infty)$ (B) $(-\infty, -2)$
(C) $(-2, 2)$ (D) $(2, \infty)$

6. Suppose f' changes from positive to negative at $x = 2$. What occurs there?

- (A) An inflection point of f (B) A vertical tangent of f
(C) A relative minimum of f (D) A relative maximum of f

7. If $f''(x) > 0$ on an interval, which statement is true?

- (A) f is increasing and f' is positive (B) f has a relative minimum everywhere
(C) f is concave down and f' is decreasing (D) f is concave up and f' is increasing

8. Find the inflection point x -coordinate of $f(x) = x^3 - 6x$.

- (A) $\sqrt{2}$ (B) $-\sqrt{2}$
(C) 0 (D) 3

Free-Response Synthesis

FRQ1 – Mean Value Theorem

Let $f(x) = x^3 - 3x$ on $[-1, 2]$.

(a) Verify the hypotheses of the MVT. [2 points]

(b) Find the secant slope. [1 points]

(c) Find all c satisfying the conclusion. [2 points]

FRQ2 – First-derivative sign chart

The derivative is $f'(x) = (x + 2)(x - 1)^2(x - 4)$.

(a) Construct a sign chart for f' . [3 points]

(b) State intervals of increase and decrease.

[2 points]

(c) Classify each critical point.

[3 points]

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